Module	Description	Example	Script
anthropic	creating a client	client = Anthropic(api_key=apikey)	g20/demo.py
anthropic	importing the module	from anthropic import Anthropic	g20/demo.py
bs4	extracting the text from a tag	pagelist = [int(a.text) for a in alist]	g33/demo.py
bs4	finding a tag with a given class	paging = soup.find(‘div’,{‘class’:‘paging’})	g33/demo.py
bs4	finding all A tags	alist = pages.find_all(‘a’)	g33/demo.py
bs4	finding an HTML title tag	title = soup.find(‘title’).text	g33/demo.py
bs4	importing the main function	from bs4 import BeautifulSoup	g33/demo.py
bs4	parsing a web page	soup = BeautifulSoup(response.text,‘html.parser’)	g33/demo.py
core	api, reading a key from a file	apikey = fh.readline().strip()	g19/demo.py
core	class, accessing the instance	cols = self.columns	g20/demo.py
core	class, defining a class attribute	sig = ‘Howdy’	g20/demo.py
core	class, defining a method	def capcols(self):	g20/demo.py
core	class, extending existing class	class myDF(pd.DataFrame):	g20/demo.py
core	continue, going on to next loop item	continue	g06/demo.py
core	dictionary, adding a new entry	co[‘po’] = ‘CO’	g05/demo.py
core	dictionary, creating	co = {‘name’:‘Colorado’, ‘capital’:‘Denver’}	g05/demo.py
core	dictionary, creating via comprehension	word_lengths = { w:len(w) for w in wordlist }	g06/demo.py
core	dictionary, deleting an element	del races[year]	g33/demo.py
core	dictionary, iterating through key-value pairs	for w,l in word_lengths.items():	g06/demo.py
core	dictionary, looking up a value	name = ny[‘name’]	g05/demo.py
core	dictionary, making a list of	list1 = [co,ny]	g05/demo.py
core	dictionary, obtaining a list of keys	names = super_dict.keys()	g05/demo.py
core	f-string, grouping with commas	print(f’Total population: {tot_pop:,}’)	g12/demo.py
core	f-string, using a formatting string	print(f”PV of {payment} with T={year} and r={r} is ${p. . .	g08/demo.py
core	file, closing	fh.close()	g02/demo.py
core	file, opening for reading	fh = open(‘states.csv’)	g05/demo.py
core	file, opening for writing	fh = open(filename,“w”)	g02/demo.py
core	file, output using print	print(“It was written during”,year,file=fh)	g02/demo.py
core	file, output using write	fh.write(“Where was this file was written?\n”)	g02/demo.py
core	file, print without adding spaces	print(‘\nOuter:\n’, join_o[’_merge’].value_counts(), s. . .	g15/demo.py

Module	Description	Example	Script
core	file, reading one line at a time	for line in fh:	g05/demo.py
core	for, looping through a list	for n in a_list:	g04/demo.py
core	for, looping through a list of tuples	for number,name in div_info:	g14/demo.py
core	function, calling	d1_ssq = sumsq(d1)	g07/demo.py
core	function, calling with an optional argument	sample_function(100, 10, r=0.07)	g08/demo.py
core	function, defining	def sumsq(values: list) -> float:	g07/demo.py
core	function, defining with optional argument	def sample_function(payment:float,year:int,r:float=0.05...	g08/demo.py
core	function, returning a result	return values	g07/demo.py
core	function, using type hinting	def readlist(filename: str) -> list:	g07/demo.py
core	if, starting a conditional block	if l == 5:	g06/demo.py
core	if, using an elif statement	elif s.isalpha():	g06/demo.py
core	if, using an else statement	else:	g06/demo.py
core	lambda function	counts = dat.apply(lambda x:len(x) if type(x)==list el...	g25/demo.py
core	lambda function	skiprows=lambda x: x in range(0,10) or x==11,	g37/demo.py
core	list, appending an element	a_list.append("four")	g03/demo.py
core	list, create via comprehension	cubes = [n**3 for n in a_list]	g04/demo.py
core	list, creating	a_list = ["zero","one","two","three"]	g03/demo.py
core	list, determining length	n = len(b_list)	g03/demo.py
core	list, extending with another list	a_list.extend(a_more)	g03/demo.py
core	list, generating a sequence	b_list = range(1,6)	g04/demo.py
core	list, joining with spaces	a_string = " ".join(a_list)	g03/demo.py
core	list, selecting an element	print(a_list[0])	g03/demo.py
core	list, selecting elements 0 to 3	print(a_list[:4])	g03/demo.py
core	list, selecting elements 1 to 2	print(a_list[1:3])	g03/demo.py
core	list, selecting elements 1 to the end	print(a_list[1:])	g03/demo.py
core	list, selecting last 3 elements	print(a_list[-3:])	g03/demo.py
core	list, selecting the last element	print(a_list[-1])	g03/demo.py
core	list, sorting	c_sort = sorted(b_list)	g03/demo.py
core	list, sorting with custom key	bylast = sorted(names, key=lambda x: x.split()[-1])	g25/demo.py
core	list, summing	total = sum(numbers)	g06/demo.py
core	math, raising a number to a power	a_cubes.append(n**3)	g04/demo.py

Module	Description	Example	Script
core	math, rounding a number	rounded = round(ratio,2)	g05/demo.py
core	sets, computing difference	print(name_states - pop_states)	g14/demo.py
core	sets, creating	name_states = set(name_data['State'])	g14/demo.py
core	sets, of tuples	tset1 = set([(1,2), (2,3), (1,3), (2,3)])	g14/demo.py
core	string, concatenating	name = s1+" "+s2+" "+s3	g02/demo.py
core	string, convert to lower case	lower = [s.lower() for s in wordlist]	g06/demo.py
core	string, convert to title case	new_s = s.title()	g06/demo.py
core	string, converting to an int	value = int(s)	g06/demo.py
core	string, creating	filename = "demo.txt"	g02/demo.py
core	string, finding starting index	mm_start = long_string.find("mm")	g06/demo.py
core	string, including a newline character	fh.write(name+"!\n")	g02/demo.py
core	string, is entirely numeric	if s.isnumeric():	g06/demo.py
core	string, matching a substring	has_ñ = [s for s in lower if "ñ" in s]	g06/demo.py
core	string, matching end	a_end = [s for s in lower if s.endswith("a")]	g06/demo.py
core	string, matching multiple starts	ab_start = [s for s in lower if s.startswith(starters)]	g06/demo.py
core	string, matching partial string	is_gas = trim['DBA Name'].str.contains('XPRESS')	g31/demo.py
core	string, matching start	a_start = [s for s in lower if s.startswith("a")]	g06/demo.py
core	string, replacing a substring	words = s.replace(","," ").split()	g06/demo.py
core	string, splitting on a comma	parts = line.split(',')	g05/demo.py
core	string, splitting on whitespace	b_list = b_string.split()	g03/demo.py
core	string, stripping blank space	clean = [item.strip() for item in parts]	g05/demo.py
core	string, using the format method	my_ask = prompt.format(tidy)	g20/demo.py
core	tuple, creating	starters = ("a","b","0")	g06/demo.py
core	type, obtaining for a variable	print('\nraw_states is a DataFrame object:', type(raw_...	g10/demo.py
core	using a try-except block	try:	g33/demo.py
core	while: looping while a condition is True	while numpage <= pagemax:	g33/demo.py
csv	setting up a DictReader object	reader = csv.DictReader(fh)	g09/demo.py
geopandas	adding a heatmap legend	slices.plot('s_pop',edgecolor='yellow',linewidth=0.2,le...	g30/demo.py
geopandas	building points from lat, lon	geom = gpd.points_from_xy(adds['lon'], adds['lat'])	g32/demo.py

Module	Description	Example	Script
geopandas	clip a layer	zips_clip = zips.clip(county,keep_geom_type=True)	g26/demo.py
geopandas	combine all geographies in a layer	water_dis = water_by_name.dissolve()	g26/demo.py
geopandas	combine geographies by attribute	water_by_name = water.dissolve('FULLNAME')	g26/demo.py
geopandas	computing areas	zips['z_area'] = zips.area	g30/demo.py
geopandas	construct a buffer	near_water = water_dis.buffer(1600)	g26/demo.py
geopandas	constructing centroids	centroids['geometry'] = tracts.centroid	g31/demo.py
geopandas	drawing a heatmap	near_wv.plot("mil",cmap='Blues',legend=True,ax=ax)	g24/demo.py
geopandas	extracting geometry from a geodataframe	wv_geo = wv['geometry']	g24/demo.py
geopandas	importing the module	import geopandas as gpd	g23/demo.py
geopandas	join to nearest object	served_by = centroids.sjoin_nearest(geo,how='left',dist...	g31/demo.py
geopandas	list layers in a geopackage	layers = gpd.list_layers(demo_file)	g26/demo.py
geopandas	merging data onto a geodataframe	conus = conus.merge(trim,on='STATEFP',how='left',valida...	g24/demo.py
geopandas	obtaining coordinates	print('Number of points:', len(wv_geo.exterior.coords)...	g24/demo.py
geopandas	overlaying a layer using union	slices = zips.overlay(county,how='union',keep_geom_type...	g30/demo.py
geopandas	plot with categorical coloring	sel.plot('NAME',cmap='Dark2',ax=ax)	g24/demo.py
geopandas	plotting a boundary	syr.boundary.plot(color='gray',linewidth=1,ax=ax)	g23/demo.py
geopandas	project a layer	county = county.to_crs(epsg=utm18n)	g26/demo.py
geopandas	reading a file	syr = gpd.read_file("tl_2016_36_place-syracuse.zip")	g23/demo.py
geopandas	reading a shapefile	states = gpd.read_file("cb_2024_us_state_500k.zip")	g24/demo.py
geopandas	reading a zip with a subdirectory	stores = gpd.read_file(zip2+'!'+subdir)	g35/demo.py
geopandas	reading data in WKT format	coords = gpd.GeoSeries.from_wkt(big['Georeference'])	g31/demo.py
geopandas	reading one layer from a zip	county = gpd.read_file(zip1+'!'+layer)	g35/demo.py
geopandas	setting the color of a plot	county.plot(color='tan',ax=ax)	g26/demo.py
geopandas	setting transparency via alpha	near_clip.plot(alpha=0.25,ax=ax)	g26/demo.py
geopandas	spatial join, contains	c_contains_z = county.sjoin(zips,how='right',predicate=...	g28/demo.py
geopandas	spatial join, crosses	i_crosses_z = inter.sjoin(zips,how='right',predicate='c...	g28/demo.py
geopandas	spatial join, intersects	z_intersect_c = zips.sjoin(county,how='left',predicate=...	g28/demo.py
geopandas	spatial join, overlaps	z_overlaps_c = zips.sjoin(county,how='left',predicate='...	g28/demo.py
geopandas	spatial join, touches	z_touch_c = zips.sjoin(county,how='left',predicate='tou...	g28/demo.py
geopandas	spatial join, within	z_within_c = zips.sjoin(county,how='left',predicate='wi...	g28/demo.py
geopandas	testing if rows touch a geometry	touches_wv = conus.touches(wv_geo)	g24/demo.py
geopandas	writing a layer to a geodatabase	conus.to_file("conus.gpkg",layer="states")	g24/demo.py
glob	importing the module	import glob	g35/demo.py
glob	listing files via wildcards	nyiso = glob.glob('raw/20??01*')	g35/demo.py
json	importing the module	import json	g05/demo.py

Module	Description	Example	Script
json	using to print an object nicely	print(json.dumps(list1,indent=4))	g05/demo.py
matplotlib	adding a y=x line	ax.axline((5,5),slope=1,color='red')	g36/demo.py
matplotlib	annotation arrow	ax.annotate('2024=2009',	g36/demo.py
matplotlib	axes, adding a horizontal line	ax21.axhline(medians['etr'], c='r', ls='–', lw=1)	g13/demo.py
matplotlib	axes, adding a vertical line	ax21.axvline(medians['inc'], c='r', ls='–', lw=1)	g13/demo.py
matplotlib	axes, labeling the X axis	ax.set_xlabel('Millions')	g12/demo.py
matplotlib	axes, labeling the Y axis	ax.set_ylabel('Millions')	g12/demo.py
matplotlib	axes, setting the left limit	ax.set_xlim(left=0)	g37/demo.py
matplotlib	axes, turning off a label	ax.set_ylabel(None)	g14/demo.py
matplotlib	axis, turning off	ax.axis('off')	g30/demo.py
matplotlib	changing marker shape	geo.plot(color='red', marker='D', markersize=20, ax=ax)	g32/demo.py
matplotlib	changing marker size	geo.plot(color='blue',markersize=1,ax=ax1)	g31/demo.py
matplotlib	colors, xkcd palette	syr.plot(color='xkcd:lightblue',ax=ax)	g23/demo.py
matplotlib	figure, adding a title	fig2.suptitle('Pooled Data')	g13/demo.py
matplotlib	figure, constrained layout	fig,ax = plt.subplots(layout='constrained')	g37/demo.py
matplotlib	figure, four panel grid	fig3, axs = plt.subplots(2,2,sharex=True,sharey=True)	g13/demo.py
matplotlib	figure, left and right panels	fig2, (ax21,ax22) = plt.subplots(1,2)	g13/demo.py
matplotlib	figure, saving	fig.savefig('figure.png')	g12/demo.py
matplotlib	figure, setting the size	fig, axs = plt.subplots(1,2,figsize=(12,6))	g21/demo.py
matplotlib	figure, tuning the layout	fig.tight_layout()	g12/demo.py
matplotlib	figure, using supxlabel	fig.supxlabel('Note: cyan=median, orange=mean',fontsize...	g37/demo.py
matplotlib	figure, working with a list of axes	for ax in axs:	g21/demo.py
matplotlib	importing pyplot	import matplotlib.pyplot as plt	g12/demo.py
matplotlib	setting a linewidth	us.boundary.plot(color='black', linewidth=0.4, ax=ax)	g32/demo.py
matplotlib	setting an axes title	ax.set_title('Hill Building Elevations')	g34/demo.py
matplotlib	setting an edge color	slices.plot('COUNTYFP',edgecolor='yellow',linewidth=0.2...	g30/demo.py
matplotlib	setting the default resolution	plt.rcParams['figure.dpi'] = 300	g12/demo.py
matplotlib	two-line labels	fig.suptitle('Distribution of Parameters\n75% and 90% e...	g37/demo.py
matplotlib	using subplots to set up a figure	fig, ax = plt.subplots()	g12/demo.py
numpy	drawing from a joint normal dist	draws_raw = np.random.multivariate_normal(est,cov,size=...	g37/demo.py
numpy	exponential function	p1_gas = np.exp(ind_vars_1['ln_p_gas'])	g37/demo.py
numpy	setting random number seed	np.random.seed(0)	g37/demo.py
os	delete a file	os.remove(out_file)	g26/demo.py
os	importing the module	import os	g19/demo.py

Module	Description	Example	Script
os	listing files in a directory	files = os.listdir('raw')	g35/demo.py
os	path, base file name	print(' Basename:', os.path.basename(fname))	g35/demo.py
os	path, directory name	print(' Dirname: ', os.path.dirname(fname))	g35/demo.py
os	path, split directory and filename	print(' Split: ', os.path.split(fname))	g35/demo.py
os	path, split filename and extension	print(' Splitext:', os.path.splitext(fname))	g35/demo.py
os	path, test for directory	print(' Dir? ', os.path.isdir(fname))	g35/demo.py
os	path, test for regular file	print(' File? ', os.path.isfile(fname))	g35/demo.py
os	test if a file or directory exists	if os.path.exists('apikey.txt'):	g19/demo.py
pandas	RE, replacing a digit or space	unit_part = values.str.replace(r'\d|\s','',regex=True)	g25/demo.py
pandas	RE, replacing a non-digit or space	value_part = values.str.replace(r'\D|\s','',regex=True)	g25/demo.py
pandas	RE, replacing a non-word character	units = units.str.replace(r'\W','',regex=True)	g25/demo.py
pandas	categorical variable, categories	print(cat.categories)	g13/demo.py
pandas	categorical variable, codes	tidy_data['cat_codes'] = cat.codes	g13/demo.py
pandas	categorical variable, creating	cat = pd.Categorical(tidy_data['type'])	g13/demo.py
pandas	columns, dividing along index	by_day_pct = 100*by_day_use.div(by_day_tot,axis='index'...	g18/demo.py
pandas	columns, dividing with explicit alignment	normed2 = 100*states.div(pa_row,axis='columns')	g10/demo.py
pandas	columns, listing names	print('\nColumns:', list(raw_states.columns))	g10/demo.py
pandas	columns, renaming	county = county.rename(columns={'B01001_001E':'pop'})	g11/demo.py
pandas	columns, retrieving one by name	pop = states['pop']	g10/demo.py
pandas	columns, retrieving several by name	print(pop[some_states]/1e6)	g10/demo.py
pandas	dataframe, appending	gen_all = pd.concat([gen_oswego, gen_onondaga])	g16/demo.py
pandas	dataframe, appending via dictionary	bg_all = pd.concat(bg_data)	g35/demo.py
pandas	dataframe, boolean row selection	print(trim[has_AM], "\n")	g13/demo.py
pandas	dataframe, checking for missing values	has_na = uns.isna().any(axis='columns')	g37/demo.py
pandas	dataframe, dropping a column	both = both.drop(columns='_merge')	g16/demo.py
pandas	dataframe, dropping duplicates	flood = flood.drop_duplicates(subset='TAX_ID')	g15/demo.py
pandas	dataframe, dropping missing data	merged = geocodes.dropna()	g12/demo.py
pandas	dataframe, finding duplicate records	dups = parcels.duplicated(subset='TAX_ID', keep=False...	g15/demo.py
pandas	dataframe, getting a block of rows via index	sel = merged.loc[number]	g14/demo.py
pandas	dataframe, inner 1:1 merge	join_i = parcels.merge(flood, how='inner', on="TAX_ID",...	g15/demo.py
pandas	dataframe, inner join	merged = name_data.merge(pop_data,left_on="State",right...	g14/demo.py
pandas	dataframe, iterating through rows	for idx,rec in sample.iterrows():	g20/demo.py
pandas	dataframe, left 1:1 merge	join_l = parcels.merge(flood, how='left', on="TAX_ID",...	g15/demo.py

Module	Description	Example	Script
pandas	dataframe, left m:1 merge	both = gen_all.merge(plants, how='left', on='Plant Code. . .	g16/demo.py
pandas	dataframe, making a copy	trim = trim.copy()	g13/demo.py
pandas	dataframe, melting	long_form = means.reset_index().melt(id_vars='month')	g18/demo.py
pandas	dataframe, outer 1:1 merge	join_o = parcels.merge(flood, how='outer', on="TAX_ID",. . .	g15/demo.py
pandas	dataframe, pivoting	by_day_use = usage.pivot(index=['month','day'],columns=. . .	g18/demo.py
pandas	dataframe, reading Excel file	wb1 = pd.read_excel(case1)	g35/demo.py
pandas	dataframe, reading Stata DTA file	wbx = pd.read_stata('bg_single.dta')	g35/demo.py
pandas	dataframe, reading missing values	na_values=missing,. . .	g37/demo.py
pandas	dataframe, reading several Excel sheets	wb2 = pd.read_excel(case2,sheet_name=None)	g35/demo.py
pandas	dataframe, reading tab-delimited data	raw = pd.read_csv(input_file,	g20/demo.py
pandas	dataframe, reading zipped pickle format	sample2 = pd.read_pickle('sample_pkl.zip')	g17/demo.py
pandas	dataframe, resetting the index	hourly = hourly.reset_index()	g18/demo.py
pandas	dataframe, right 1:1 merge	join_r = parcels.merge(flood, how='right', on="TAX_ID",. . .	g15/demo.py
pandas	dataframe, saving in zipped pickle format	sample.to_pickle('sample_pkl.zip')	g17/demo.py
pandas	dataframe, selecting rows by list indexing	print(low_to_high[-5:])	g10/demo.py
pandas	dataframe, selecting rows via boolean	dup_rec = flood[dups]	g15/demo.py
pandas	dataframe, selecting rows via query	trimmed = county.query("state =='04' or state == '36' ")	g11/demo.py
pandas	dataframe, selective drop of missing data	trim = demo.dropna(subset="Days")	g13/demo.py
pandas	dataframe, set index keeping the column	states = states.set_index('STUSPS',drop=False)	g24/demo.py
pandas	dataframe, shape attribute	print('number of rows, columns:', conus.shape)	g24/demo.py
pandas	dataframe, skipping rows for excel files	g_raw = pd.read_excel(. . .	g37/demo.py
pandas	dataframe, sorting by a column	county = county.sort_values('pop')	g11/demo.py
pandas	dataframe, sorting by index	summary = summary.sort_index(ascending=False)	g16/demo.py
pandas	dataframe, stacking	stack = allcases.stack().reset_index(1)	g37/demo.py
pandas	dataframe, summing a boolean	print('\nduplicate parcels:', dups.sum())	g15/demo.py
pandas	dataframe, summing across columns	by_day_tot = by_day_use.sum(axis='columns')	g18/demo.py
pandas	dataframe, unstacking an index level	bymo = bymo.unstack('month')	g18/demo.py
pandas	dataframe, using a multilevel column index	means = grid['mean']	g21/demo.py
pandas	dataframe, using compare	cmp = wb_a.compare(wb_b,result_names=('wb_a','wb_b'))	g35/demo.py
pandas	dataframe, using xs to select a subset	print(county.xs('04',level='state'))	g11/demo.py
pandas	dataframe, using xs with columns	c1 = grid.xs('c1',axis='columns',level=1)	g21/demo.py
pandas	dataframe, writing Stata DTA file	wb1.to_stata('bg_single.dta',write_index=False)	g35/demo.py
pandas	dataframe, writing as an Excel file	trim.to_excel(output_file,index=False)	g20/demo.py
pandas	dataframe, writing to a CSV file	merged.to_csv('demo-merged.csv')	g14/demo.py
pandas	datetime, building via to_datetime()	date = pd.to_datetime(recs['ts'])	g15/demo.py
pandas	datetime, building with a format	ymd = pd.to_datetime(sample['TRANSACTION_DT'], format=. . .	g17/demo.py

Module	Description	Example	Script
pandas	datetime, extracting day attribute	recs['day'] = date.dt.day	g15/demo.py
pandas	datetime, extracting hour attribute	recs['hour'] = date.dt.hour	g15/demo.py
pandas	general, describe with percentiles	pct_stats = pct.describe(percentiles=[0.05,0.95])	g37/demo.py
pandas	general, display information about object	sample.info()	g17/demo.py
pandas	general, displaying all columns	pd.set_option('display.max_columns',None)	g17/demo.py
pandas	general, displaying all rows	pd.set_option('display.max_rows', None)	g10/demo.py
pandas	general, importing the module	import pandas as pd	g10/demo.py
pandas	general, using copy_on_write mode	pd.options.mode.copy_on_write = True	g17/demo.py
pandas	general, using qcut to create deciles	dec = pd.qcut(county['pop'], 10, labels=range(1,11))	g11/demo.py
pandas	groupby, cumulative sum within group	cumulative_inc = group_by_state['pop'].cumsum()	g11/demo.py
pandas	groupby, descriptive statistics	inc_stats = group_by_state['pop'].describe()	g11/demo.py
pandas	groupby, iterating over groups	for t,g in group_by_state:	g11/demo.py
pandas	groupby, median of each group	pop_med = group_by_state['pop'].median()	g11/demo.py
pandas	groupby, quantile of each group	pop_25th = group_by_state['pop'].quantile(0.25)	g11/demo.py
pandas	groupby, return group number	groups = group_by_state.ngroup()	g11/demo.py
pandas	groupby, return number within group	seqnum = group_by_state.cumcount()	g11/demo.py
pandas	groupby, return rank within group	rank_pop = group_by_state['pop'].rank()	g11/demo.py
pandas	groupby, select first records	first2 = group_by_state.head(2)	g11/demo.py
pandas	groupby, select largest values	largest = group_by_state['pop'].nlargest(2)	g11/demo.py
pandas	groupby, select last records	last2 = group_by_state.tail(2)	g11/demo.py
pandas	groupby, size of each group	num_rows = group_by_state.size()	g11/demo.py
pandas	groupby, sum of each group	state = county.groupby('state')['pop'].sum()	g11/demo.py
pandas	index, creating with 3 levels	county = county.set_index(['state','county', 'NAME'])	g11/demo.py
pandas	index, dropping a level	races.index = races.index.droplevel('title')	g33/demo.py
pandas	index, listing names	print('\nIndex (rows):', list(raw_states.index))	g10/demo.py
pandas	index, renaming values	div_pop = div_pop.rename(index=div_names)	g12/demo.py
pandas	index, retrieving a row by name	pa_row = states.loc['Pennsylvania']	g10/demo.py
pandas	index, retrieving first rows by location	print(low_to_high.iloc[0:10])	g10/demo.py
pandas	index, retrieving last rows by location	print(low_to_high.iloc[-5:])	g10/demo.py
pandas	index, set name of index level	bgs_all.index = bgs_all.index.set_names('year',level=0)	g36/demo.py
pandas	index, setting to a column	states = raw_states.set_index('name')	g10/demo.py
pandas	plotting, bar plot	reg_pop.plot.bar(title='Population',ax=ax)	g12/demo.py
pandas	plotting, histogram	hh_data['etr'].plot.hist(ax=ax1,bins=20,title='Distribu. . .	g13/demo.py

Module	Description	Example	Script
pandas	plotting, horizontal bar plot	div_pop.plot.barh(title=‘Population’,ax=ax)	g12/demo.py
pandas	plotting, scatter colored by 3rd var	tidy_data.plot.scatter(ax=ax4,x=‘Income’,y=‘ETR’,c=’Reg...	g13/demo.py
pandas	plotting, scatter plot	hh_data.plot.scatter(ax=ax21,x=‘inc’,y=‘etr’,title=’ETR...	g13/demo.py
pandas	plotting, stacked bar plot	races.plot.bar(stacked=True,ax=ax)	g33/demo.py
pandas	plotting, turning off legend	sel.plot.barh(x=‘Name’,y=‘percent’,ax=ax,legend=None)	g14/demo.py
pandas	reading, csv data	raw_states = pd.read_csv(‘state-data.csv’)	g10/demo.py
pandas	reading, from an open file handle	gen_oswego = pd.read_csv(fh1)	g16/demo.py
pandas	reading, setting index column	state_data = pd.read_csv(‘state-data.csv’,index_col=’na...	g12/demo.py
pandas	reading, using dtype dictionary	county = pd.read_csv(‘county_pop.csv’,dtype=fips)	g11/demo.py
pandas	series, RE at start	is_LD = trim[‘Number’].str.contains(r”$^{1|}$2”)	g13/demo.py
pandas	series, applying a function to each element	name_clean = name_parts.apply(’ ’.join)	g25/demo.py
pandas	series, automatic alignment by index	merged[‘percent’] = 100*merged[‘pop’]/div_pop	g14/demo.py
pandas	series, check if all elements are true	print((wb1 == wbx).all())	g35/demo.py
pandas	series, combining via mask()	mod[‘comb_mask’] = unit_part.mask(unit_part==” , mod[...	g25/demo.py
pandas	series, combining via where()	mod[‘comb_where’] = unit_part.where(unit_part!=” , mo...	g25/demo.py
pandas	series, computing natural logs	res[f’ln_{v}’] = res[v].apply(np.log)	g37/demo.py
pandas	series, contains RE or RE	is_TT = trim[‘Days’].str.contains(r”Tu|Th”)	g13/demo.py
pandas	series, contains a plain string	has_AM = trim[‘Time’].str.contains(“AM”)	g13/demo.py
pandas	series, contains an RE	has_AMPM = trim[‘Time’].str.contains(“AM.*PM”)	g13/demo.py
pandas	series, converting strings to title case	fixname = subset_view[‘NAME’].str.title()	g17/demo.py
pandas	series, converting to a list	print(name_data[‘State’].to_list())	g14/demo.py
pandas	series, converting to lower case	name = mod[‘name’].str.lower()	g25/demo.py
pandas	series, dropping rows using a list	conus = states.drop(not_conus)	g24/demo.py
pandas	series, element-by-element or	is_either = is_ca | is_tx	g17/demo.py
pandas	series, filling missing values	mod[‘comb_units’] = mod[‘comb_where’].fillna(‘feet’)	g25/demo.py
pandas	series, random sample	vins = vins.sample(frac=0.001,random_state=789)	g20/demo.py
pandas	series, removing spaces	units = units.str.strip()	g25/demo.py
pandas	series, replacing values using a dictionary	units = units.replace(spellout)	g25/demo.py
pandas	series, retrieving an element	print(“\nFlorida’s population:”, pop[‘Florida’]/1e6)	g10/demo.py
pandas	series, sort in decending order	div_pop = div_pop.sort_values(ascending=False)	g12/demo.py
pandas	series, sorting by value	low_to_high = normed[‘med_pers_inc’].sort_values()	g10/demo.py
pandas	series, splitting strings on whitespace	name_parts = name.str.split()	g25/demo.py
pandas	series, splitting via RE	trim[‘Split’] = trim[“Time”].str.split(r”:| - | “)	g13/demo.py
pandas	series, splitting with expand	exp = trim[“Time”].str.split(r”:| - | “, expand=True)	g13/demo.py
pandas	series, summing	reg_pop = by_reg[‘pop’].sum()/1e6	g12/demo.py

Module	Description	Example	Script
pandas	series, unstacking	tot_wide = tot_amt.unstack('PGI')	g17/demo.py
pandas	series, using isin()	fixed = flood['TAX_ID'].isin(dup_rec['TAX_ID'])	g15/demo.py
pandas	series, using value_counts()	print('\nOuter:\n', join_o['_merge'].value_counts(), s...	g15/demo.py
pydantic	defining an object	class ComplaintInfo(BaseModel):	g20/demo.py
pydantic	importing the module	from pydantic import BaseModel	g20/demo.py
rasterio	checking compression	if ras.compression is not None:	g34/demo.py
rasterio	checking the resolution	print(f' width = {ras.res[0]}')	g34/demo.py
rasterio	count of bands	print('Bands:',ras.count)	g34/demo.py
rasterio	importing the module	import rasterio	g34/demo.py
rasterio	obtaining the CRS	dem_crs = hill_ras.crs	g34/demo.py
rasterio	opening a raster	hill_ras = rasterio.open(hill_dem_file)	g34/demo.py
rasterio	retrieving band description	print('Description:',ras.descriptions[band-1])	g34/demo.py
rasterstats	computing categorical statistics	nlcd_stats = zonal_stats(nys,	g34/demo.py
rasterstats	computing zonal statistics	hill_stats = zonal_stats(hill_fp,	g34/demo.py
rasterstats	importing zonal_stats	from rasterstats import zonal_stats	g34/demo.py
requests	adding headers	response = requests.get(api,payload,headers=headers)	g32/demo.py
requests	calling the get() method	response = requests.get(api,payload)	g19/demo.py
requests	checking the URL	print('url:', response.url)	g19/demo.py
requests	checking the response text	print(response.text)	g19/demo.py
requests	checking the status code	print('status:', response.status_code)	g19/demo.py
requests	decoding a JSON response	rows = response.json()	g19/demo.py
requests	geocoding via nominatim	api = "https://nominatim.openstreetmap.org/search"	g32/demo.py
requests	getting a web page	response = requests.get(base_url,payload)	g33/demo.py
requests	importing the module	import requests	g19/demo.py
scipy	calling newton's method	cr = opt.newton(find_cube_root,xinit,maxiter=20,args=[y...	g08/demo.py
scipy	importing the module	import scipy.optimize as opt	g08/demo.py
seaborn	adding a title to a grid object	jg.fig.suptitle('Distribution of Hourly Load')	g18/demo.py
seaborn	barplot	hue='month',palette='deep',ax=ax1)	g18/demo.py
seaborn	basic violin plot	sns.violinplot(data=janjul,x="month",y="usage")	g18/demo.py
seaborn	boxenplot	sns.boxenplot(data=janjul,x="month",y="usage")	g18/demo.py
seaborn	calling tight_layout on a grid object	jg.fig.tight_layout()	g18/demo.py

Module	Description	Example	Script
seaborn	confidence ellipsoid	sns.kdeplot(...	g37/demo.py
seaborn	displot	g = sns.displot(data=stack,x='pct',hue='Case',kind='his...	g37/demo.py
seaborn	drawing a heatmapped grid	sns.heatmap(means,annot=True,fmt=".0f",cmap='Spectral',...	g21/demo.py
seaborn	importing the module	import seaborn as sns	g18/demo.py
seaborn	joint distribution hex plot	jg = sns.jointplot(data=bymo,x=1,y=7,kind='hex')	g18/demo.py
seaborn	line plot	sns.lineplot(data=long_form,x='hour',y='value',hue='mon...	g18/demo.py
seaborn	setting axis titles on a grid object	jg.set_axis_labels('January','July')	g18/demo.py
seaborn	setting the theme	sns.set_theme(style="white")	g18/demo.py
seaborn	split violin plot	hue="month",palette='deep',split=True)	g18/demo.py
sql	IN clause using a subquery	WHERE PID IN (SELECT ...)	g27/demo.py
sql	appending to a table via pandas	n = df.to_sql('enrollment',con,if_exists='append',index...	g29/demo.py
sql	changing data type with CAST	CAST('Grid kV' AS NUMERIC) AS kV	g27/demo.py
sql	connecting to a SQLite database	con = sqlite3.connect(eia_file)	g27/demo.py
sql	counting records	COUNT(*) AS count	g27/demo.py
sql	create table with primary key	cur = con.execute(create_table)	g29/demo.py
sql	creating a table with a unique constraint	cur = con.executescript(create_enrollment)	g29/demo.py
sql	creating a view	CREATE VIEW summary AS SELECT ...	g29/demo.py
sql	deleteing records	cur = con.execute(f"DELETE FROM races WHERE year={year}...	g33/demo.py
sql	dropping a table	DROP TABLE IF EXISTS courses	g29/demo.py
sql	executing a SQL script	cur = con.executescript(sql_cmds)	g29/demo.py
sql	insert using a template	INSERT INTO courses VALUES (?,?,?)	g29/demo.py
sql	inserting a single record	cur = con.execute(insert_one)	g29/demo.py
sql	inserting multiple rows via executemany	cur = con.executemany("INSERT INTO courses VALUES (?,?,...	g29/demo.py
sql	obtaining a list of tables	SELECT name,sql FROM sqlite_master	g27/demo.py
sql	reading a table via pandas	utility = pd.read_sql("SELECT * FROM Utility;", con)	g27/demo.py
sql	retrieving column names from cursor	cur_info = cur.description	g27/demo.py
sql	retrieving count of rows affected	print('\nRows affected',cur.rowcount)	g29/demo.py
sql	retrieving rows via fetchall	rows = cur.fetchall()	g27/demo.py
sql	select column name with spaces	SELECT ... 'Energy Source'	g27/demo.py
sql	setting join keys with USING	JOIN ... USING(UID)	g27/demo.py
sql	simple select of all columns	cur = con.execute("SELECT * FROM Utility;")	g27/demo.py
sql	sort in descending order	ORDER BY total_mw DESC	g27/demo.py
sql	starting a with block	with con:	g29/demo.py
sql	summing a variable	SUM('Nameplate MW') AS mw	g27/demo.py
sql	updating fields for selected records	cur = con.execute(update_cmd)	g29/demo.py
sql	using DISTINCT	cur = con.execute("SELECT DISTINCT year FROM races;")	g33/demo.py

Module	Description	Example	Script
sql	using LIKE in a WHERE clause	WHERE Name LIKE '%national grid%'	g27/demo.py
sql	using fetchone to retrieve one row	check = cur.fetchone()	g29/demo.py
sql	using set in an update	SET name = REPLACE(name,'Intro','Introduction')	g29/demo.py
sql	where with an IN clause	WHERE p.State IN ('VT','NH')	g27/demo.py
statsmodels	get parameter covariance matrix	cov = results.cov_params()	g37/demo.py
statsmodels	get parameters	est = results.params	g37/demo.py
statsmodels	importing	import statsmodels.api as sm	g37/demo.py
statsmodels	including a constant	X = sm.add_constant(X)	g37/demo.py
statsmodels	regression summary	print(results.summary())	g37/demo.py
statsmodels	running a regression	results = model.fit()	g37/demo.py
statsmodels	setting dependent variable	Y = res[dep_var]	g37/demo.py
statsmodels	setting independent variables	X = res[ind_vars]	g37/demo.py
statsmodels	setting up an OLS model	model = sm.OLS(Y,X)	g37/demo.py
stringio	accessing a string as a file	tables = pd.read_html(StringIO(response.text))	g33/demo.py
stringio	importing the function	from io import StringIO	g33/demo.py
sys	exiting a script	sys.exit()	g20/demo.py
sys	importing the module	import sys	g20/demo.py
tomllib	importing the module	import tomllib	g36/demo.py
tomllib	loading the file	setup = tomllib.load(fh)	g36/demo.py
tomllib	opening the file	with open(setup_file,'rb') as fh:	g36/demo.py
zipfile	importing the module	import zipfile	g16/demo.py
zipfile	opening a file in an archive	fh1 = archive.open('generators-oswego.csv')	g16/demo.py
zipfile	opening an archive	archive = zipfile.ZipFile('generators.zip')	g16/demo.py
zipfile	reading the list of files	print(archive.namelist())	g16/demo.py